Number System Conversion Table

Decimal, binary, octal and hexadecimal - values 0 to 63

DEC	BINARY	OCT	HEX
1	000001	001	01
3	000011	003	03
5	000101	005	05
7	000111	007	07
9	001001	011	09
11	001011	013	0B
13	001101	015	0D
15	001111	017	0F
17	010001	021	11
19	010011	023	13
21	010101	025	15
23	010111	027	17
25	011001	031	19
27	011011	033	1B
29	011101	035	1D
31	011111	037	1F

DEC	BINARY	OCT	HEX
33	100001	041	21
35	100011	043	23
37	100101	045	25
39	100111	047	27
41	101001	051	29
43	101011	053	2B
45	101101	055	2D
47	101111	057	2F
49	110001	061	31
51	110011	063	33
53	110101	065	35
55	110111	067	37
57	111001	071	39
59	111011	073	3B
61	111101	075	3D
63	111111	077	3F

Powers of two

2^0 = 1
2^1 = 2
2^2 = 4
2^3 = 8
2^4 = 16
2^5 = 32
2^6 = 64
2^7 = 128
2^8 = 256
2^9 = 512
2^10 = 1,024
2^11 = 2,048
2^12 = 4,096
2^13 = 8,192
2^14 = 16,384
2^15 = 32,768
2^16 = 65,536
2^20 = 1,048,576
2^24 = 16,777,216
2^30 = 1,073,741,824
2^32 = 4,294,967,296
2^40 = 1,099,511,627,776

Read a row across: the same value in all four bases. Hex digits A-F represent 10-15.

Methods and World Scales

How to convert, and how 10 languages name 1 million, 1 billion and 1 trillion

Method 1 - decimal to another base (repeated division)

1. Divide the number by the target base and record the remainder.
2. Divide the quotient again by the base. Repeat until the quotient is 0.
3. Read the remainders from bottom to top - that is the converted number.

Example: 156 / 2 = 78 r0, 78/2=39 r0, 39/2=19 r1, 19/2=9 r1, 9/2=4 r1, 4/2=2 r0, 2/2=1 r0, 1/2=0 r1 -> 10011100.

Method 2 - any base to decimal (positional weights)

1. Number the digit positions from 0 at the right of the integer part, going left.
2. Multiply each digit by (base ^ position) and add the results.

Example: 1A.8 in hex = 1x16^1 + 10x16^0 + 8x16^-1 = 16 + 10 + 0.5 = 26.5.

Fractions: multiply the fraction by the base, take the carry, repeat. Read carries top to bottom.

Indian and International scale

POWER	INTERNATIONAL	INTL FORMAT	INDIAN	INDIAN FORMAT
10^3	Thousand	1,000	Thousand	1,000
10^5	Hundred thousand	100,000	Lakh	1,00,000
10^6	Million	1,000,000	Ten lakh	10,00,000
10^7	Ten million	10,000,000	Crore	1,00,00,000
10^9	Billion	1,000,000,000	Arab	1,00,00,00,000
10^11	Hundred billion	100,000,000,000	Kharab	1,00,00,00,00,000
10^12	Trillion	1,000,000,000,000	Lakh crore	1,00,000 crore
10^15	Quadrillion	1,000,000,000,000,000	Padma	1,00,00,00,00,00,00,000

Where "billion" changes meaning

English / International	short	billion = 10^9	trillion = 10^12
Indian	indian	arab = 10^9 (100 crore)	lakh crore = 10^12
Portuguese (Portugal)	long	mil milhoes = 10^9	biliao = 10^12
Portuguese (Brazil)	short	bilhao = 10^9	trilhao = 10^12
Spanish	long	mil millones = 10^9	billon = 10^12
French	long	milliard = 10^9	billion = 10^12
German	long	Milliarde = 10^9	Billion = 10^12
Japanese / Chinese / Korean	myriad	10 oku / 10 yi = 10^9	cho / zhao / jo = 10^12
Russian, Turkish, Arabic	short	milliard/milyar = 10^9	trillion = 10^12

Portuguese: Portugal keeps the long scale, Brazil uses the short scale - check the locale before converting.

Source and more worked examples: numbersystemconverter.com/knowledge-catalog/